Project Proposal: Multivariate Analysis of Dementia Patient Health Data

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Background and Aim

This project will analyze the **Dementia Patient Health Dataset** available on Kaggle, which contains demographic information, lifestyle characteristics, physiological measurements, cognitive indicators, medical history, and dementia status.

The main aim of the project is to explore the multidimensional structure of the dataset and investigate whether combinations of patient characteristics are associated with dementia. Both unsupervised and supervised multivariate statistical methods will be applied in order to identify underlying patterns, patient groups, and variables related to dementia status.

Research Questions

1. **What are the main variables underlying the demographic, clinical, cognitive, and lifestyle characteristics of the patients?**
2. **Can patients be grouped into distinct health profiles without using dementia diagnosis as an input variable?**
3. **Do the naturally identified patient clusters correspond to the observed dementia status?**
4. **Which variables are significantly associated with the presence of dementia?**
5. **Can groups of related variables explain different aspects of variability among patients?**

Proposed Methodology

- **Principal Component Analysis (PCA)**
- **Exploratory Factor Analysis (EFA)**
- **Clustering methods** (K-means)
- **Logistic regression**

Expected Outcome

The project is expected to provide a comprehensive multivariate description of the dataset, identify major patterns and latent dimensions in patient health characteristics, reveal potential patient subgroups, and determine which variables are most strongly associated with dementia status.